Distance Weight-Graph Attention Model-Based High-Resolution Remote Sensing Urban Functional Zone Identification

Kui Zhang, Dongping Ming, Shigao Du, Lu Xu, Xiao Ling, Beichen Zeng, and Xianwei Lv

***Abstract*— The spatial arrangement of land-cover features constitutes different urban functional zone. With the same attributes of the urban functional zone, the land-cover features that make up the functional zone will have similar spatial distribution characteristics. Considering the importance of understanding spatial relationships between land-cover features, the up-bottom hierarchical decomposition and semantic understanding of functional zone are achieved. First, for object convolution neural network (OCNN)-based land cover classification, an equal-area dividing algorithm is proposed to automatically generate convolution kernel position. Second, to extract spatial relationship features of urban land covers, a novel distance weight-graph attention model (DW-GAM) is originally proposed for classifying urban functional zones by comparing the feature similarity of the land cover relationship graph. Third, considering the extreme difficulties in expressing the urban structure characteristic on a single scale, a recursive model that uses an urban road network of different levels to divide multiscale functional zones is built. Finally, taking the analysis of urban function allocation as the application objective, this article establishes a primary framework of the spatial pattern evaluation index. Experimental results conducted on a Google Earth image of Xi'an city show that the multiscale recursive model can accurately recognize urban functional zones by using the originally proposed DW-GAM. Then, based on the outcome of urban functional zone identification, the case study of urban function allocation analysis is innovatively conducted on the fine scale to give some suggestions on future urban planning, which is, hence, of great significance for urban function pattern analysis and urban planning.**

***Index Terms*— Distance weight-graph attention model (DW-GAM), multiscale recursive model, urban function pattern, urban functional zone.**

NOMENCLATURE

DW-GAM	Distance weight-graph attention model.
OCNN	Object convolution neural network.
VHSR	Very high spatial resolution.
FD	Frequency density.
CR	Category ratio.
OBIA	Object-based image analysis.
ENN	Euclidean nearest neighbor distance.
PLAND	Percentage of the landscape.
AWMPFD	Area-weighted mean patch fractal dimension.
CONTACT	Contagion index.
DIVISION	Landscape division index.
SHDI	Shannon's diversity index.
SHEI	Shannon's evenness index.

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I. INTRODUCTION

AS A part of the urban complicated organism, the urban functional zone is a geographical space that performs the urban function, and it is embodied by the aggregation of relevant social resources. Urban functional zones with the same attributes often have the same requirements on the types and spatial allocations of land cover. However, due to the different spatial distributions of land cover, some human activities usually gather in the same local space; thus, various urban functional zones are formed. Therefore, accurately identifying urban land cover and the spatial relationship among land cover is helpful for modeling and spatial division of the urban functional zone.

VHSR remote sensing imagery can provide rich surface information of the Earth, which makes it possible to extract urban land cover features accurately. The extraction accuracy of land cover features directly affects urban functional zones' identification [1]. However, the spectral differences of the same land cover become larger, the spectral characteristics of different land cover are more similar, and traditional classification methods based on low- and medium-resolution images cannot be directly used for VHSR image classification [2]–[4]. Therefore, the OBIA method is proposed and widely used in VHSR images, which greatly improves the classification accuracy of land covers [5], [6]. Many studies applied OBIA in urban land cover classification with a set of low-level features (such as spectra, texture, and shape) of the ground features; however, it is difficult to express the deep-level features of land cover [7].

In order to fully express the complex features of land cover, the feature representation framework based on deep learning is widely used in VHSR images classification [8]–[10]; the convolutional neural network (CNN) is one of the important representative models of deep learning method. Many studies

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classified remote sensing images by OCNN [11], [12]. The OCNN method can extract the deep features of the image, including the global information, and keep the spatial detailed information [13]–[17].

The key of the OCNN model is how to select convolution position in the segmented object as the convolution position affects classification accuracy. There are two main sampling methods: Zhang *et al.* [18] proposed the object convolutional position analysis (OCPA) method, which is mainly aimed at the linearly shaped objects; Lv *et al.* [12] proposed the random point generation algorithm, but the spatial distribution of random points is uneven. The reasonable distribution of convolution position is helpful to improve urban land cover classification accuracy, and it lays the groundwork for identifying urban functional zone.

Recently, studies on extending graph-based deep learning approaches have emerged in natural language understanding. These topological graph data are completely consistent with the spatial relationship between land cover features in the functional zone, which casts a light on urban functional zone identification. Graph-based deep learning models have been applied to a variety of analysis tasks, including the most common node classification [19], connection edge prediction [20]–[22], and graph classification [23]–[25]. However, graph classification is still in the exploratory stage due to the complexity of the own method and its application though it is good at extracting spatial relation features; thus, applying graph classification in urban functional zone identification will be a promising way because the graph structure existing within the urban function zone can be effectively used.

In addition, the urban functional pattern is closely related to the distribution of urban functional zone. The combination and interaction of the different functional zones determine the urban functional pattern, and the rationality of the urban functional pattern restricts urban functional zones formation [26], [27]. Reasonable planning of the functional pattern provides a strong basis for urban sustainable development [28], [29]. Current research on functional patterns mainly focuses on the allocation analysis of large-scale and multitemporal [30]–[32], which ignores the fine-scale functional pattern analysis. Therefore, it is valuable work to extract the functional zone on the fine scale and build up a fine-scale evaluation index system for urban functional allocation analysis and then further put forward reasonable suggestions for meeting urban functional configuration requirements.

Based on the above analysis, this article proposes an object-based graph classification model, DW-GAM, which focuses on extracting spatial relation features between different objects to further extract multiscale functional zones by jointly using a multilevel road network and point of interest (POI) data with VHSR image. Especially, this article establishes the systematic framework for urban functional pattern evaluation, which is further verified by a case study of urban functional zone identification and allocation analysis.

The remainder of this article is organized as follows. Section II introduces the related work on the identification of urban functional zones. Section III introduces the general workflow and the key components of the proposed methods. Section IV describes the study area and experiment results, followed by a discussion in Section V. The conclusions are drawn in Section VI.

II. Related Work on Identification of Urban Functional Zones

There are two main steps in urban functional zone identification: urban functional zone division and category classification [33].

A. Urban Functional Zone Division

Urban functional zone division methods can be subdivided into single-scale division methods and multiscale division methods. There are actually two approaches to single-scale division: dividing methods based on remote sensing image or vector auxiliary data. The first method uses image segmentation technology to generate urban functional zone units according to the spatial distribution features of image objects and the homogeneity of functional types [34]. Zhou *et al.* [35] proposed that the image was segmented into superobjects (SOs) by using single-scale division. The second method relies on vector auxiliary data, such as road network and grid, and the road network is also used to divide functional zones at a single scale. However, there is no regulation on which level of the road should be selected as the standard of urban functional zone division [36]–[38].

Du *et al.* [39] achieved the multiscale division method through the scale-adaptive approach and graph cuts algorithm. The method does not use vector auxiliary data, especially road networks, which is very important for reflecting and evaluating urban functional patterns. In order to effectively express the hierarchical semantic information of urban functional zones, this article will explore the significance of multiscale functional zones division based on road network data.

B. Urban Functional Zone Classification

For the attribute recognition of urban functional zones, the existing recognition algorithms mainly include clustering algorithms and deep learning. The clustering algorithm is mainly aimed at socially perception data with geographical location [40]–[42], such as POI. POIs reflect functional semantic information and reveal the spatiotemporal behavior patterns of human beings within the city. Therefore, urban functional zones can be directly analyzed from social perception data. However, when there is no POI in the block unit, these methods cannot determine the attribute of the functional zone.

The deep learning model, with the ability to extract the deep features of the urban functional zone, can be applied to identify the functional zone. The CNN model can mine the deep image features and spatial information, which significantly improves the accuracy and calculation efficiency of the functional zone attribute recognition [43]–[45]. CNN can excavate the topological relationship characteristics between pixels within a specific window but cannot express the spatial adjacency relation between land cover features on the object level. However, the spatial attribute characteristics between

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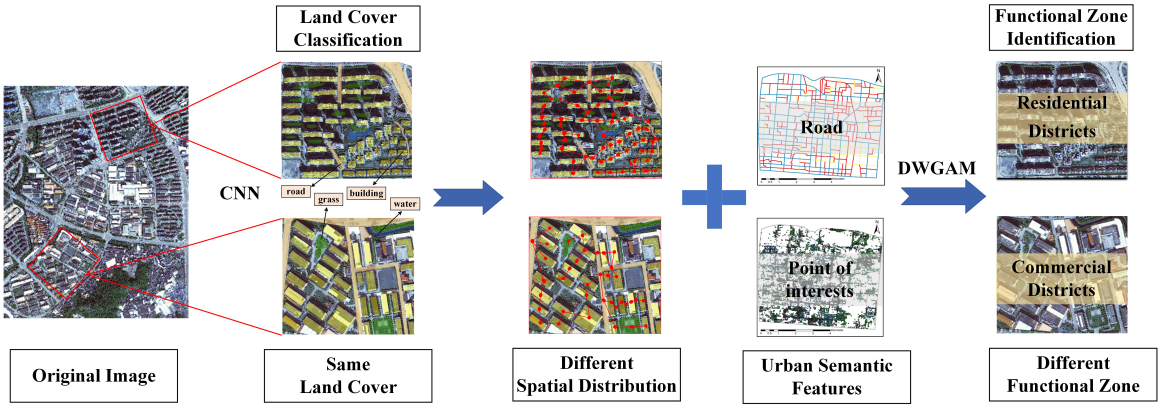

Fig. 1. Workflow of urban functional zones' identification.

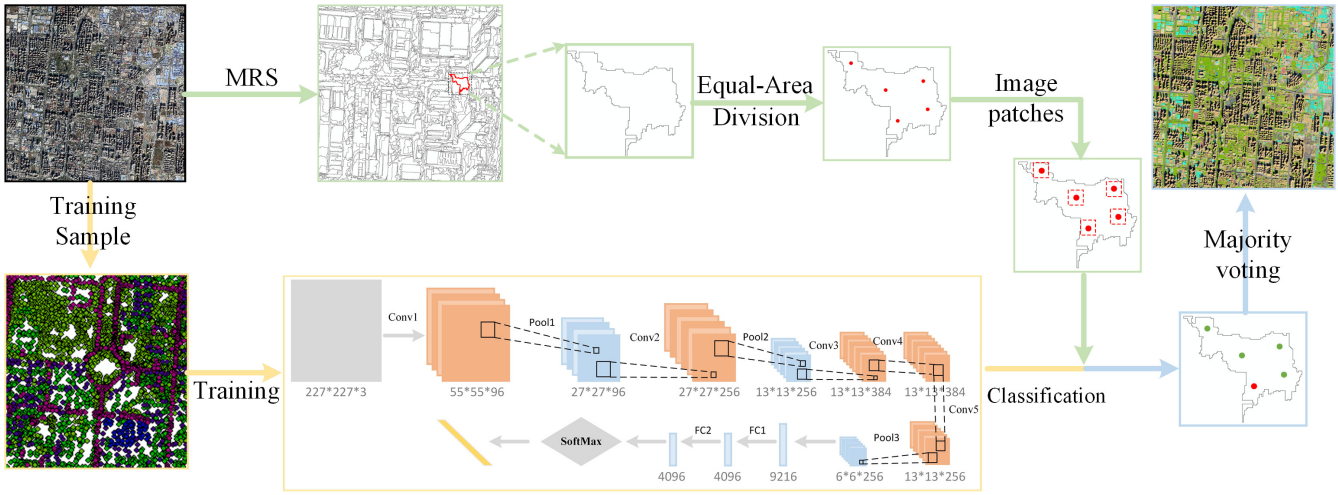

Fig. 2. Flowchart of the OCNN method (the green points (p1, p2, p3, and p5) represent the correctly classified category points; one red point (p4) represents misclassified points).

adjacent land cover greatly influence the functional zone identification. Constructing a machine learning model capable of mining the spatial relationship features between land covers is the key to urban functional zone recognition.

Due to the deficiencies of currently functional zone division and classification methods, the spatial features between land cover are extracted by the deep learning model, and the hidden social semantic features of POI are excavated. More details about the principle and methodology for functional zone dividing and identifying are presented in Section III.

III. Methodology

Fig. 1 illustrates the workflow of urban functional zones' identification, which includes the following three steps.

Step 1 (Object-Based Land Cover Classification): VHSR image is segmented into image objects, and then, the CNN model based on the proposed equal-area division algorithm is used to classify the image object to obtain the land cover classification results of the study area.

Step 2 (Top-Down Functional Zone Identification): The spatial distribution features of each block unit are extracted by DW-GAM, and then, combined with the urban semantic features represented by POI, functional zone categories of multiscale block units can be finely identified level by level.

Step 3: Urban functional allocation analysis based on the self-established assessment index system.

A. Equal-Area Division Algorithm

In this article, the OCNN method combining OBIA and CNN is introduced for urban land cover classification. As shown in Fig. 2, the OCNN classification method comprises the following three steps. First, the training samples are manually marked to train the parameter values of the OCNN model according to Google map and VHSR images. Second, the VHSR image is segmented into image objects by the segmentation method, and each object is processed by an equal-area division algorithm, in order to generate convolution positions with reasonable spatial distribution inside the object. Finally, given an object, image patches are generated at each convolution position as test samples, and each patch is classified by the pretrained model. The predicted category of each convolution position may be inconsistent, and the majority voting strategy is used to determine the attribute of the object (the final attribute of the object is determined by the largest number of predicted categories).

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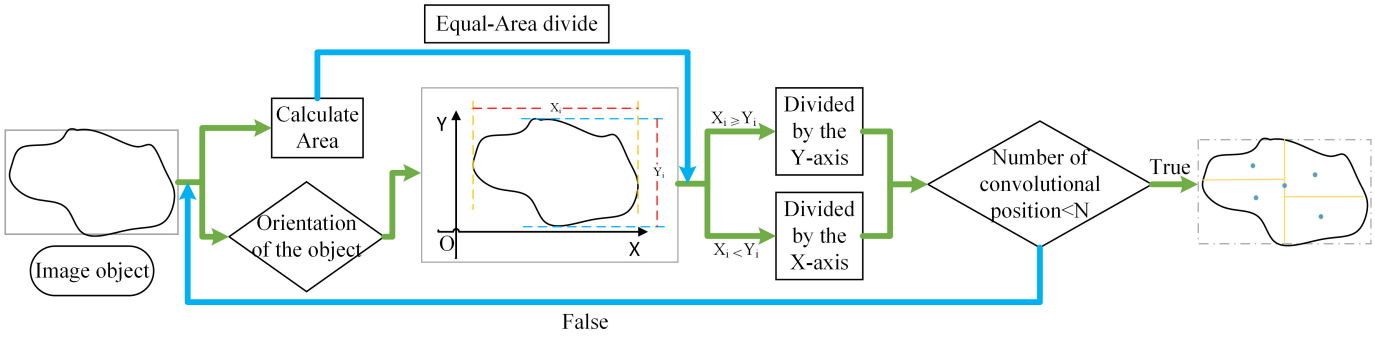

Fig. 3. Workflow of the equal-area division algorithm.

The selection of convolution kernel position affects the classification accuracy of OCNN. The normally used random point generation algorithm proposed for the RMV-CNN model is suitable for objects of various shapes, but the convolution kernel position may fall on the object edge, which cannot accurately extract the object features.

The OCPA was proposed based on the moment bounding (MB) box of each object to identify the position, by which the convolution position is distributed on both sides of the MB major axis. However, an idealized convolution position should meet the following requirements. First, it must be evenly and reasonably distributed among each object. Second, the algorithm for generating convolution positions should adapt to an image object with any shape.

Thus, this article proposes an iterative sampling strategy based on the idea of equally dividing the object area. The algorithm realizes the object hierarchical division according to the object area, as shown in Fig. 3. Before dividing the secondary objects, it is necessary to judge whether the objects are distributed horizontally or vertically in order to make the subdivided object's shape relatively consistent. Each iteration will generate equal-area subdivided objects (one divides into two) until the specified even number is generated. Finally, the centroid points of all subdivided objects are generated; thus, all convolution position points with odd numbers (including the centroid point of the original image object) are generated. Fig. 3 illustrates the generated convolutional kernel positions (the points with blue color) of OCNN for the case of $N = 5$. The equal-area division method is applied only to the testing phase in the study.

B. DW-GAM-Based Urban Functional Zone Identification

1) DW-GAM: The graph-based deep learning model has made some achievements in chemistry, medicine, and other fields. Compared with other deep learning models (such as CNN), it can learn the topological relationship between nodes and nontopological attribute features (such as distance and azimuth) between adjacent nodes. Learning the deep-level spatial distribution features between land covers is necessary for classifying urban functional zone because different urban functional zone presents specific spatial structures. Therefore, this article attempts to introduce an RNN-based deep learning model for graph classification. The novel RNN model, called

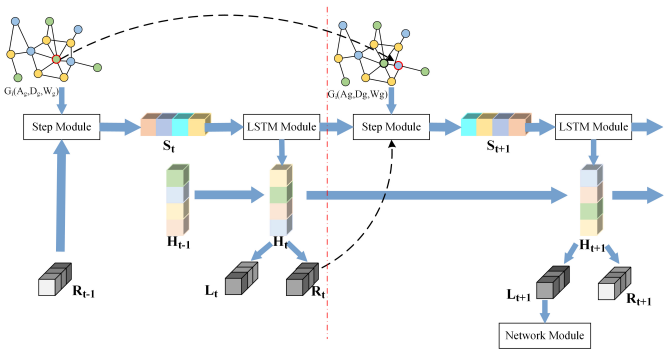

Fig. 4. Structure of DW-GAM.

the graph attention model (GAM) [46], predicts graph labels by traversing the nodes in part of the graph, which is involved in urban functional zone identification for the first time.

For identification of urban functional zone, the two factors should be taken into account: there is a strong spatial autocorrelation between land cover features in the urban functional zone; features with spatial topological structure should satisfy the first law of geography, the distance is an essential factor for the correlation between land covers, and the closer the distance between land covers, the greater the correlation. Therefore, by adding the distance weight to the GAM-based step module, this article proposes the DW-GAM architecture, as shown in Fig. 4.

DW-GAM consists of three parts: step module, LSTM module, and network module. The input of step module is each topology data $G_i(\mathbf{A_g}, \mathbf{D_g}, \mathbf{W_g}) = \{g_i, l_i\}$($g_i$ refers to the ith graph and is comprised of an adjacency matrix $\mathbf{A_g}$, attribute matrix $\mathbf{D_g}$, distance weight matrix $\mathbf{W_g}$, and the rank vector r_{t-1} (a stochastic rank vector, prioritizing those whose type (i.e., node label have a higher rank in r_{t-1}); l_i is graph label in the ith graph). Through training and learning, the output of the step model is the embedding vector (s_t), which represents information captured from the current step that we took.

The information vector s_t output by step module, as the input vector of the LSTM module, will update the historical vector h_t. h_t is the latest feature vector after the representation of graph features, and the model will constantly update the feature information over time. h_t can be defined as

$$h_t = y_h(s_t, h_{t-1}; \theta_h) \tag{1}$$

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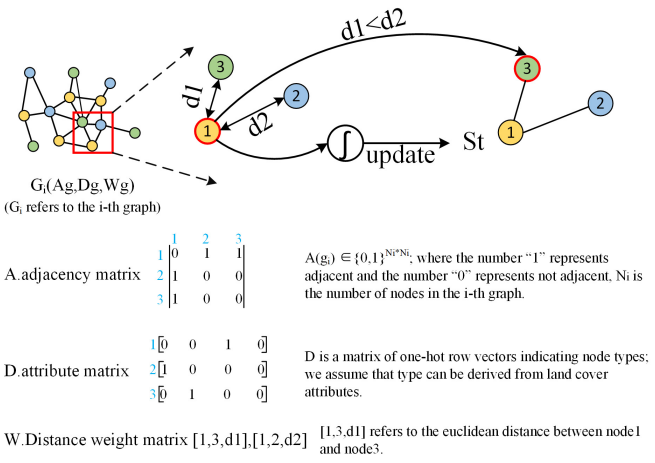

Fig. 5. Schematic of the step module.

where h_{t-1} is the historical vector generated at the previous time, which is equivalent to a feature vector of short-term memory, and θ_h is the training parameter.

The newly updated history vector h_t has two functions. First, a new rank vector r_t is defined as $r_t = y_r(h_t; \theta_r)$, and r_t will be used as the input vector for the next step module. Second, h_t is used as the input vector of the network module. The network module is designed to predict graph label (l_t) with the help of the latest history vector h_t. The function of predicting graph labels can be written as $\widehat{l_t} = \arg\max_i P(y = i | y_l(h_t; \theta_l))$.

The main goal of the step module is to select the next node to be moved. The method of traversal nodes used in the step module of GAM is to find out the type of node that occurs the most frequently among adjacent nodes. In this article, the calculated distance weights are input into the step module to enable DW-GAM to find the nearest neighboring node among adjacent nodes.

Taking the case of only three nodes as an example, Fig. 5 illustrates the workflow of the node traversal. First, the main land covers from the VHSR image classification results in each block unit are extracted. Then, the land covers in each block unit are treated as nodes of the graph data, calculating the adjacency matrix (in which the number "1" represents adjacent and the number "0" represents not adjacent), attribute matrix (recording the land cover type of each node, for example, the land cover type of node 1 is 0110), and distance weight matrix (in which d1 denotes the spatial distance from node 1 to node 3, and d2 denotes the spatial distance from node 1 to node 2) for the nodes in each block unit. The distance weight matrix is to find the nearest node to the current node. Second, the three matrixes are input into the step module, and the spatial structure features of urban land covers can be acquired by learning the spatial distribution patterns among different land covers and then generate the new node output vectors of the module.

Compared with the computational process of GAM, the distance weight is the time-consuming stage for DW-GAM. The GAM gives rise to a time complexity of $O(N)$, where N represents the number of neighbor nodes. DW-GAM needs to find the nearest node by traversing the neighbor nodes again. Each node has an acceptable number of neighbor nodes, so it does not have a significant impact on the performance of the model.

2) Urban Functional Zone Classification: Previous studies of determining urban functional zone categories are mainly relying on POI, DMSP/OLS nighttime light image, GPS trajectory, and other multisource data to recognize urban functional attributes [47]–[50]. Studies have extracted urban semantic features of POI data to classify urban functional zone through clustering algorithms or mathematical formulas [51], [52]. Some research methods recognize functional zone by integrating remote sensing images and POI data [53]–[55]. Most previous studies only relied on the urban functional semantics of multisource data [56] and, however, failed to mine the spatial relationship features between land covers in VHSR images [57], [58].

The urban functional zone contains multiple land cover categories with different spectral and spatial features. Sometimes, the same land cover may exist in a different urban functional zone; however, the spatial relationships or spatial arrangement of land covers is different. Given the land cover diversity in VHSR images and land cover spatial distribution variability, it is necessary to build a more robust feature learning model, which can learn the spatial structure relations among land covers and, thus, contribute to accurate functional zone attribute identification. Therefore, DW-GAM is selected as the main model framework for the attribute identification of urban functional zones.

POIs are introduced as auxiliary data for urban functional zone classification. POI data contain a large amount of urban functional semantic information, which can be used to quantitatively identify the urban functional zone. However, the number of POIs contained in each functional zone is not evenly distributed [59]. For example, the amount of POIs of commercial types is significantly more than other types, and even no distribution of POI data appears in some blocks. Moreover, it is difficult to guarantee the accuracy of the POI attributes in some blocks, so it is not feasible to only rely on POI to recognize urban functional zone because open access unstructured urban data can be biased since there is uncertainty around the data's representativeness [60]. In this article, the quantitative identification method of POI data is another identification method of functional zones. We compare the consistency of urban functional zone identification results between the DW-GAM method and POI data.

Considering that the number of POIs within different functional attributes is different, we identify functional zone attributes by constructing FD and CR for a block unit according to formulas (2) and (3)

$$\mathrm{FD}_i = \frac{n_i}{N_i} \quad (i = 1, 2, 3, 4, 5) \tag{2}$$

$$\mathrm{CR}_i = \frac{\mathrm{FD}_i}{\sum_{i=1}^{5} \mathrm{FD}_i} \quad (i = 1, 2, 3, 4, 5) \tag{3}$$

where i refers to the ith type of POI in the block unit; n_i is the total number of the ith type of POI in the block unit;

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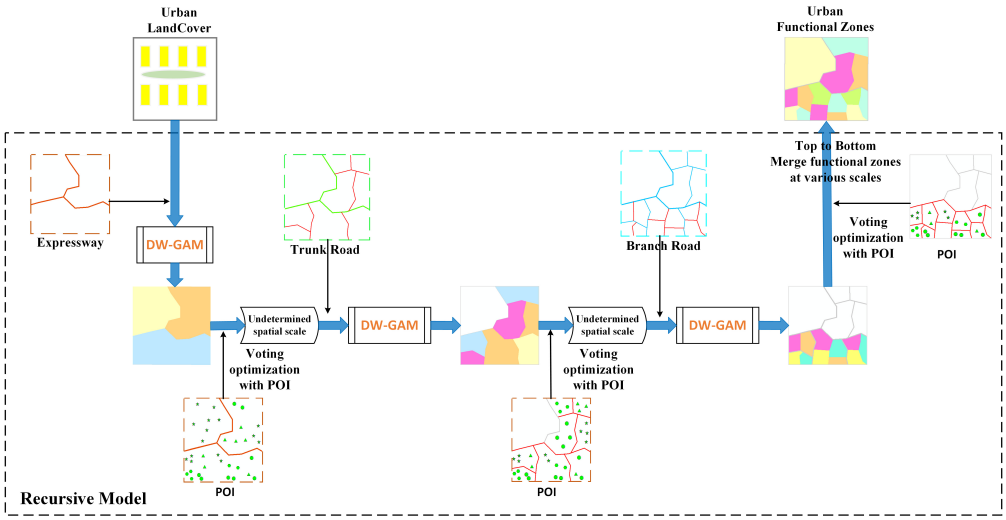

Fig. 6. Multiscale recursive model of functional zone identification.

N_i is the total number of all POI in the block unit; F_i refers to the FD of the ith type of POI in the block unit; and C_i is the CR of the ith type of POI in the block unit. The CR value of 50% is recommended as the standard to judge the functional categories of each block unit.

Meanwhile, considering the multilevel road network provided the feasibility of fine extraction of the urban functional zone, a multiscale recursive model of functional zone identification was proposed by combining DW-GAM and POI quantitative methods. Fig. 6 presents the multiscale recursive model of functional zone identification. We divide the block units using road network data from the large scale to the small scale. We use two methods to distinguish the attributes of each block unit: DW-GAM and POI quantitative method [formula (2) and formula (3)]. Correspondingly, DW-GAM is employed to determine the urban functional zone categories of block units on each scale. In this procedure, POI-based voting is used to optimize the DW-GAM-based recognition, and the voting results determine which blocks will be divided into subblocks on the next scale. This recursive model can recognize urban functional zones for block units on different scales.

3) Multiscale Urban Functional Zone Division: A variety of auxiliary data are used for the division of urban functional zones, such as grid data [61], cadastral data [62], [63], and road network data [38], [64], among which block unit formed by road network is the most widely used in functional zone division because it is closest to the boundary of the urban functional zone and easy to be acquired [36], [37], [65].

The urban functional zone division method based on road network takes the actual urban road distribution into consideration; however, a spatially and semantically independent functional zone may be overdivided in the road-dense area, which increases the complexity of division. Therefore, a multiscale functional zone division method is proposed by using multilevel urban road network data, which decreases the computation of generating block units.

After calculating all CRs for all POI types within each block unit, the block units with CR value less than 50% will be further divided by the next level road network until the new CR value is more than 50%. Then, the urban functional zone attribute can be determined by calculating the attribute similarity between CR-based judgment and DW-GAM-based recognition. As illustrated in Algorithm 1, the rules for determining multiscale functional zone division are given as follows.

Algorithm 1 Multiscale Functional Zone Division Algorithm

1: Input: r_i, r_{i+1}, **M**, POI (r_i is road of i-th scale, **M** is DW-GAM identification result of i-th scale)
2: Initialize: correct_graph (an empty list to store geospatial unit ID without next scale division)
question_graph (an empty list to store geospatial unit ID require next scale division)
question_featureclass(a new featureclass to store next scale road)
3: Process:
4: for j = 0 to n do (n is the total number of road levels)
5: for k = 0 to p do (p is the block unit number of j-th scale)
6: Calculate **C[j][k]** and **M[j][k]**(**C[j][k]** is CR value of k-th block unit at j-th scale, **M[j][k]** is DW-GAM result of k-th block unit at j-th scale)
7: if (Max(**C[j][k]**) $\geq$ 0.5 && $\mathbf{M_j}$ == F(Max(**C[j][k]**)) or Max(**C[j][k]**) == null) then
8: append(**M[j][k]**) to correct_graph
9: else append(**M[j][k]**) to question_graph
10: question_graph generate question_featureclass
11: question_featureclass clip r_{i+1} generate new_r, and as a new model input (new_r is the graph data generated under the new scale)
12: Output: new_r

1) For the block units with a CR value of more than 50%, if the functional attribute determined by CR is consistent with the functional identification result by DW-GAM,

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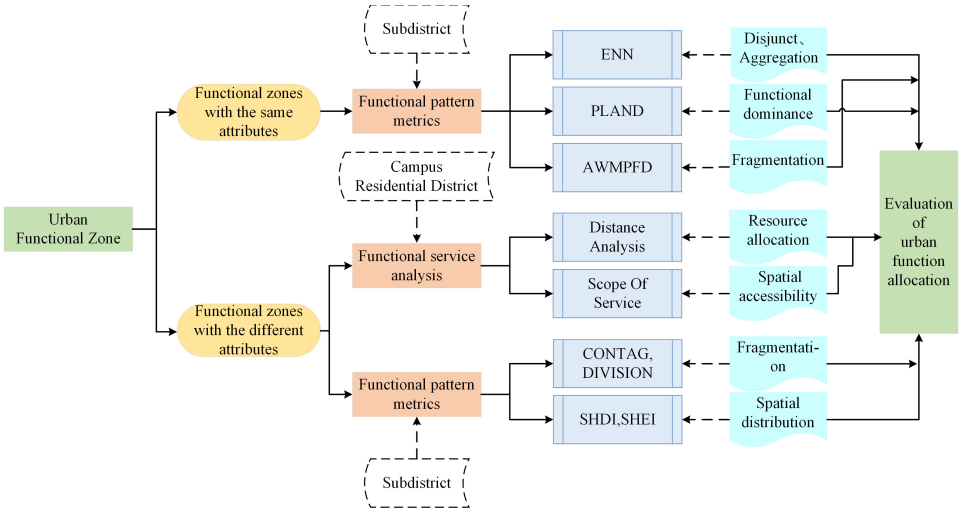

Fig. 7. Framework of urban function allocation analysis (ENN: Euclidean nearest neighbor distance; PLAND: percentage of the landscape; AWMPFD: area-weighted mean patch fractal dimension; CONTACT: contagion index; DIVISION: landscape division index; SHDI: Shannon's diversity index; and SHEI: Shannon's evenness index).

the functional zone attribute of the block unit will be easily determined, and this block unit will not be divided at the next scale.

2) For the block units with a CR value of more than 50%, if the functional attribute determined by CR is inconsistent with the functional identification result by DW-GAM, the block unit will be further divided by the next level road network until the achieved two functional attributes are consistent.
3) For the block unit does not contain POI (CR is a null value), the block unit is called a null value unit, the identification result of DW-GAM will be the terminal functional zone category of the block unit, and the block unit also will not be divided at the next scale.

As shown in Fig. 6, the DW-GAM identification results and the calculation results of CR are combined. The partition of the next scale of block units depends on whether the identification results of the two methods are consistent, so as to realize the top-down hierarchical division from the large-scale road network to small-scale road network. On the one hand, this method completes the mutual verification of the two identification results, which ensures the extraction accuracy of the functional zone; on the other hand, it reduces the unnecessary partition of some blocks, which greatly reduces the number of DW-GAM graph datasets and improves the operation efficiency of the model.

C. Evaluation System of Urban Function Pattern Allocation

Urban functional distribution in urban space forms the urban spatial structure that also reflects the spatial manifestation of the city's society and culture in the region [66]. Therefore, the spatial pattern of urban functions determines the social and cultural development direction of the city, and a reasonable function configuration is important to promote urban sustainable development [28], [67].

Urban function configuration research is based on rough scale urban pattern analysis, which results in that the analysis is consequently rough. Based on the urban functional zone identification results from VHSR images, enlightened by landscape pattern analysis, this article establishes the evaluation index of urban functional pattern, as shown in Fig. 7. A series of indexes of functional pattern and functional service is established from two perspectives: the spatial distribution of the same attribute functional zone and the spatial correlation between different attribute functional zones. By using the established evaluation index of urban functional patterns, it is available to realize the preliminary functional allocation analysis and provide rational urban planning suggestions for future urban development.

IV. Experiment

The experiments were performed on an Ubuntu 16.04 OS with a CPU (3.4 GHz core i7-6700), RAM (8 GB), and GPU (NVIDIA GTX 1060 2 GB). PyTorch was chosen as the deep learning framework.

A. Experimental Data

As shown in Fig. 8(a), the experimental data, with a spatial resolution of 0.6 m and a size of 12 246 × 11 002 pixels, were downloaded from Google Map. The image was captured on July 21, 2019, and was located in the urban area of Weiyang district, Xi'an city, China.

Two images, Fig. 8(b) and (c), with a spatial resolution of 0.6 m and sizes of 9320 × 12 427 pixels and 13 502 × 9527 pixels, respectively, were used to generate samples of graph dataset for DW-GAM.

POI data record the activity types of urban residents in a certain place, including living, working, dining, shopping, or entertainment. We took 32 840 POIs as the vector auxiliary data [see Fig. 8(d)], which were downloaded from the Baidu map and divided into five categories: residential services, commercial services, education services, scenic spots, and shantytowns.

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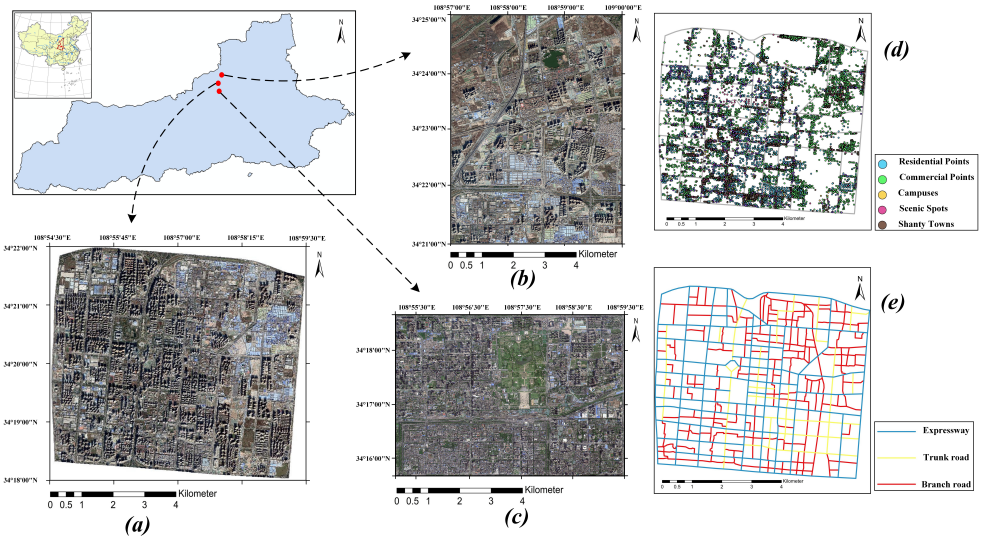

Fig. 8. Experimental data. (a) VHSR image for model predicting. (b) and (c) VHSR image for model training. (d) POIs. (e) OpenStreetMap (OSM).

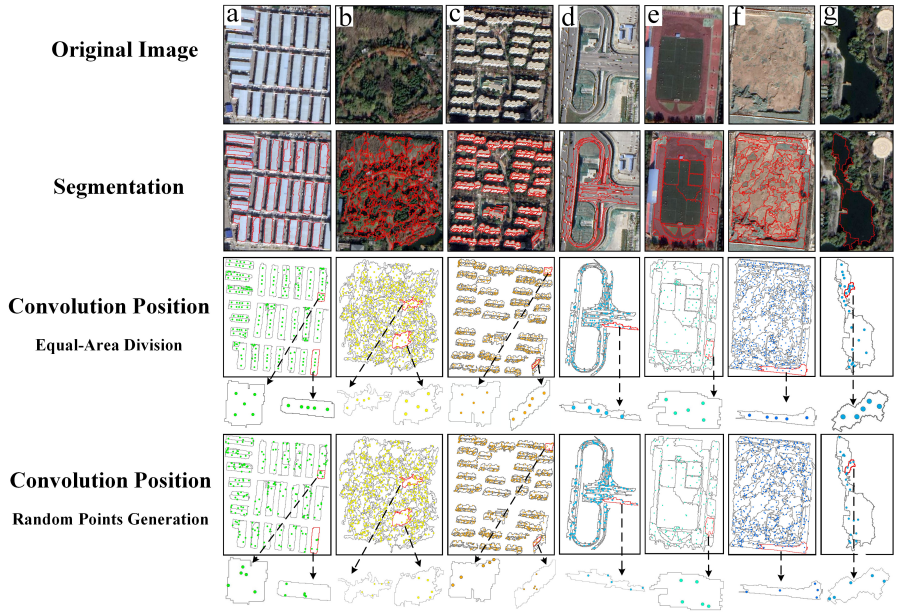

Fig. 9. Local results of convolution position. (a) Building with the blue roof. (b) Grass. (c) Other building. (d) Road. (e) Artificial turf field. (f) Bare soil. (g) Water.

Fig. 8(e) showed the vector data of the three-level road network obtained from the open street map (OSM), including expressways, trunk roads, and branch roads.

B. Urban Land Cover Classification Based on Equal-Area Division Algorithm

1) Equal-Area Division Algorithm: The proposed equal-area division algorithm is sensitive to the segmented regions. In order to acquire a better segmented boundary, we chose the multiresolution segmentation (MRS) algorithm. The segmentation parameters were determined by experience. For the Xi'an image, the scale parameter was set as 30, while the shape and compactness values were set as 0.2 and 0.4, respectively.

Based on the segmentation results, positions of convolution for each object were generated by an equal-area division strategy. As the equal-area division results cannot be shown clearly here, this article only demonstrated part of the results of convolutional positions. Fig. 9 showed the local results of the equal-area division algorithm and random points' generation algorithm, respectively.

Fig. 9 showed the local results of seven different land covers. There were some problems in the convolutional positions extracted by the random point generation algorithm. For example, some points were too close, and some points were distributed on the edge of the object, so CNN was limited to mine local features of the image object. The points generated by the equal-area division algorithm were suitable

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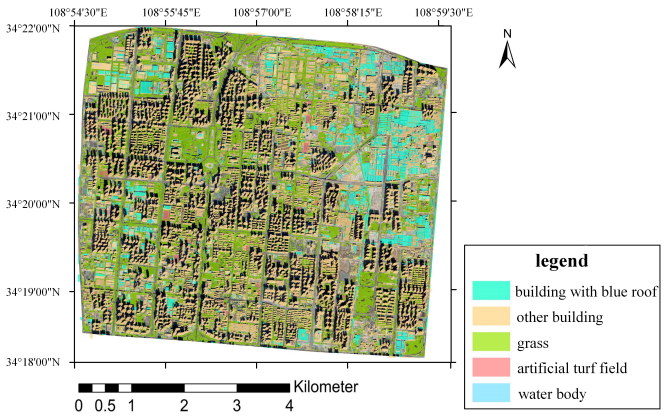

Fig. 10. Urban land cover classification result.

for objects of different shapes and ensured that they were evenly distributed in the objects. Therefore, the features of each object can be extracted more completely by CNN, which will improve the classification accuracy to a certain extent.

2) Urban Land Cover Classification: AlexNet, a fundamental CNN framework, was introduced to classify urban land cover. Some parameters need to be clearly explained. For training the CNN model, the learning rate was set as 0.01 to control the progress of the learning model. The epoch is the number of times that entire training datasets were traversed by the model. Here, the epoch of the framework this article used was set to 100. Avoiding overfitting of the deep learning model, the dropout rate was set as 0.5. The batch is a part of the training datasets fed into the model each time, and the batch size is the number of training samples in each batch. We set the batch size to 100.

As shown in Fig. 10, after sorting out the classification results, five land cover features, namely, building with blue roofs, other buildings, grass, artificial turf field, and water body, were selected to identify urban functional zones.

C. Identification of Urban Functional Zones Based on DW-GAM

1) Training Data Sampling: In this article, the categories of urban functional zones are determined according to the code for classification of urban land use and planning standards of development land, which is issued by the Ministry of Housing and Urban-Rural Development of the People's Republic of China. A label sample is composed of an attribute matrix, adjacency matrix, weight matrix, and its functional label. The training samples of each category were selected by means of POI data and online Baidu Map (https://map.baidu.com). To enhance the robustness of the DW-GAM, we made training samples by random sampling from Fig. 8(b) and (c), 183 graph datasets (including 23 816 nodes) were generated for training the graph classification model, and the sample sizes of the commercial district, campus, residential district, scenic spot, and shanty down are 45, 16, 87, 15, and 20, respectively. Fig. 11 illustrates representative samples (image patch) of each functional zone category.

2) Parameters Setting for DW-GAM: In order to extract the deep spatial features of each graph, DW-GAM is needed to set different parameters, such as CNN. The learning rate that

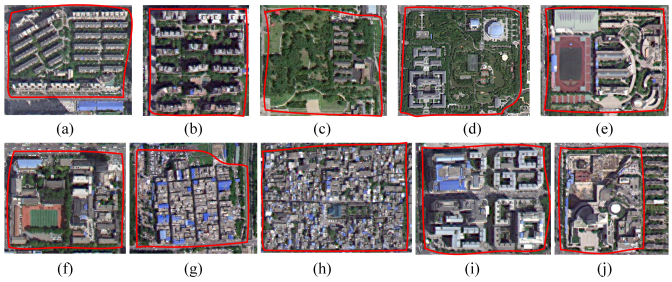

Fig. 11. Selected ten representative samples. (a) and (b) Residential district. (c) and (d) Scenic spot. (e) and (f) Campus. (g) and (h) Shanty down. (i) and (j) Commercial district.

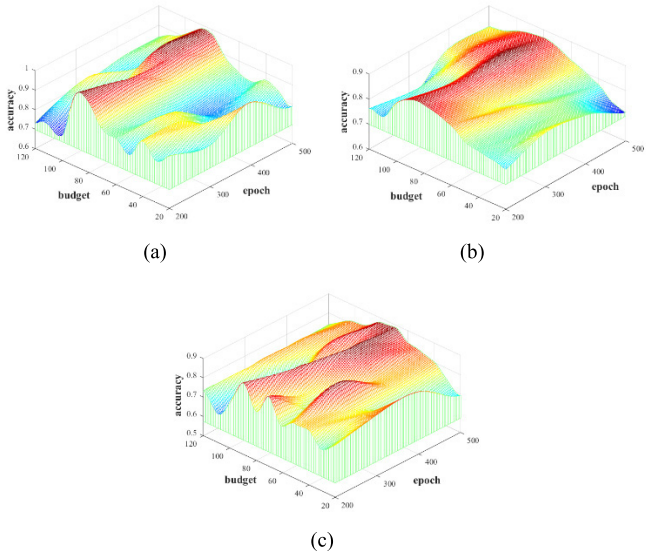

Fig. 12. Changes of different scales with the budget values varying from 20 to 120 and the epochs varying from 200 to 500. (a) Expressway scale. (b) Trunk road scale. (c) Branch road scale.

controlled the progress of DW-GAM was 0.01. In addition, through a large number of experiments, it was concluded that epoch and budget have a great influence on the experimental results. The size of the epoch is to extract the deep spatial features of graphs and prevent overfitting. The budget determines the number of nodes that be traversed in each epoch. If the number of traversed nodes is small, DW-GAM cannot completely mine the deep-level features of the graph. Rather, if the number of traversed nodes is large, the model will learn the features of some fuzzy nodes, which will have an inappropriate impact on the classification accuracy.

For the multiscale functional zones' identification method, due to the inconsistency of the size of the graphs at different scales, the amount of nodes contained in each graph is different, but the size of the graphs at the same scale is relatively consistent. We made an analysis on budget and epoch at three scales. According to the results in Fig. 12, the values of the budget and epoch parameters corresponding to the expressways, trunk roads, and branch roads were set as 100 and 500; 80 and 400; and 70 and 400, respectively.

3) Multiscale Urban Functional Zone Identification Results: Scale is the main issue in the urban functional zone identification. Functional zones with the same attributes may have block units of different scales. This is because of the

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Fig. 13. Functional zone identification results at the expressway scale.

Fig. 14. Functional zone identification results at the trunk road scale.

Fig. 15. Functional zone identification results under the branch road scale.

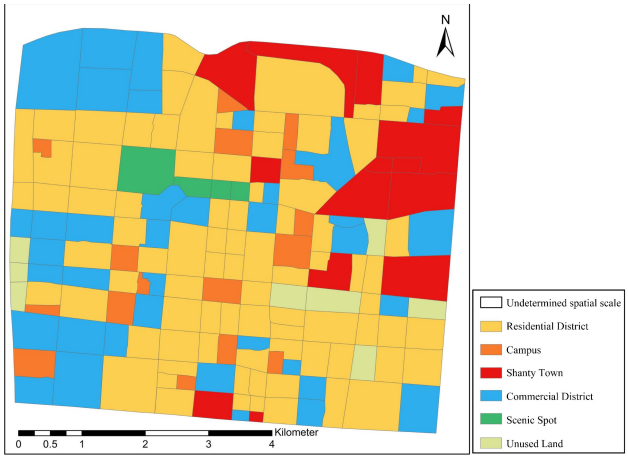

Fig. 16. Functional zones' identification results.

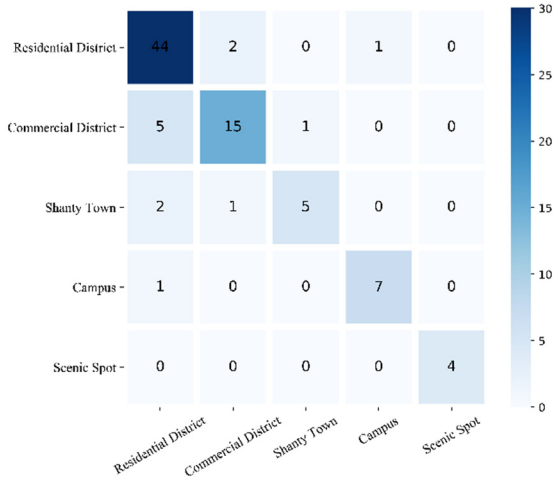

Fig. 17. Confusion matrix.

difference in urban planning and design. For example, some residential districts are bounded by the expressways, and these residential districts are relatively large in scale compared with the residential districts bounded by branch roads. Therefore, we adopted a multiscale functional zones' identification method to find the best division scale for each functional zone as far as possible and to ensure that the functional attribute features and spatial features of one block unit are relatively homogeneous.

Fig. 13 showed the recognition results of functional zones with expressway as the first level of division scale. The expressway is mainly responsible for the long-distance and rapid transport services in the city; the division scale is relatively large. The study area was divided into 58 first-level block units, of which only 18 block units were successfully identified by the multiscale model, and the other 40 block units failed to reach the threshold of the similarity calculation. It is shown that the functional attributes of these 40 block units were highly heterogeneous. There were multiple categories of functional zones inside the block units, and these first-level block units are needed to be segmented at the next scale.

Taking the trunk road as the dividing boundary, a total of 88 second-level block units were divided, of which 51 second-level block units were successfully identified. The remaining 37 second-level block units are needed to be divided into the next scale (see Fig. 14). Most of the residential districts in the second-level block units were identified, which

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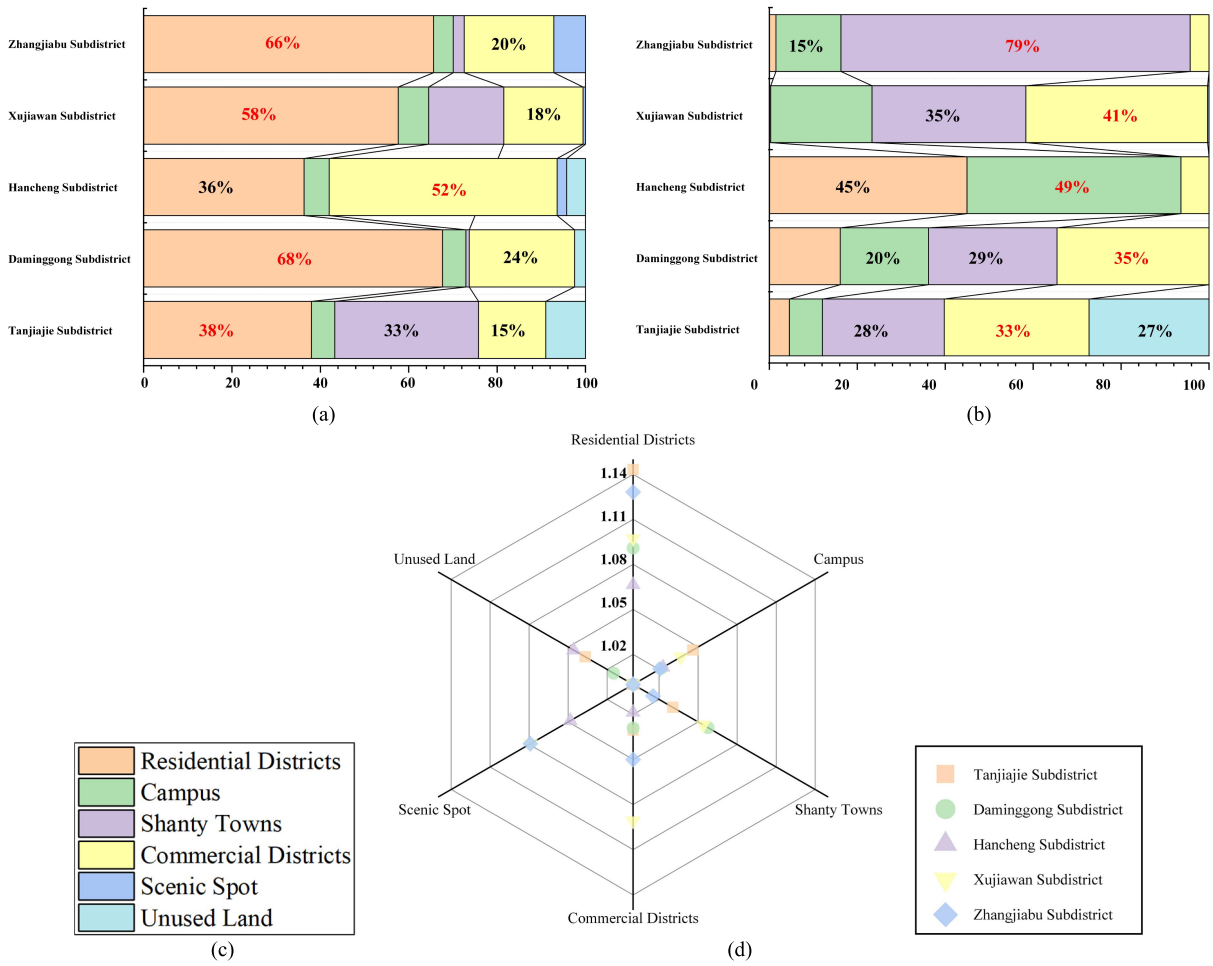

Fig. 18. Urban function pattern index results. (a) PLAND. (b) ENN. (c) Legend of Fig.17(a) and (b). (d) AWMPFD.

also indicated that many residential districts in urban planning were designed with the trunk roads as the boundary. In addition, the trunk roads with traffic functions are used to connect the main districts of the city, and residential districts are generally located near streets with convenient traffic.

Fig. 15 showed the recognition results of the last-level block unit divided by branch roads. There were 101 third-level block units identified by DW-GAM, and commercial districts and campuses were identified at the smallest scale. Commercial districts mainly contained some companies and factories. The area of the functional zones was relatively small. Therefore, the smallest scale branch roads were required to divide the functional zones' units.

Finally, we combined the classification results of the above three scales to obtain the overall urban functional zones' identification results of the study area, as shown in Fig. 16. The proposed algorithm not only considered the heterogeneity of land covers in the block units but also divided the urban functional zones from the large-scale roads network to the small-scale roads network and realized the classification of multiscale urban functional zones from top to bottom. A total of 170 block units in the study area were classified. For each category of urban functional zones, about 40% of block units of the classification results were extracted, and the results were tested with the true value of the Google electronic vector map. The confusion matrix was calculated, as shown in Fig. 17, and the overall classification accuracy was 85.2%.

D. Analysis of Allocation for Urban Function Pattern

For the purpose of analyzing the allocation of urban function reasonably and elaborately, the study area was subdivided into five blocks; therefore, the evaluation of spatial distribution that functional subdistrict corresponding to each block was derived.

1) Allocation Analysis of the Same Functional Zones: Each subdistrict owns its urban function distinctively, equipped with corresponding fundamental functional zones, and, meanwhile, has its representative domination functional zones as well-developed tourism zones would locate in some certain subdistricts while advanced commercial zone in another. Therefore, function pattern indexes were calculated in this section for the purpose of analyzing the aggregation degree of a total of five subdistricts along with their domine attributes.

The index PLAND represents the proportion of landscape types, which was calculated for analyzing regional dominant urban function; Fig. 18(a) shows that, except for the Hancheng subdistrict, the proportion of residential district appeared to be the highest in any other zones. Tanjiajie subdistrict locates in the east of the study area where a large number of factories exist as well, and the shantytown proportion rises relatively.

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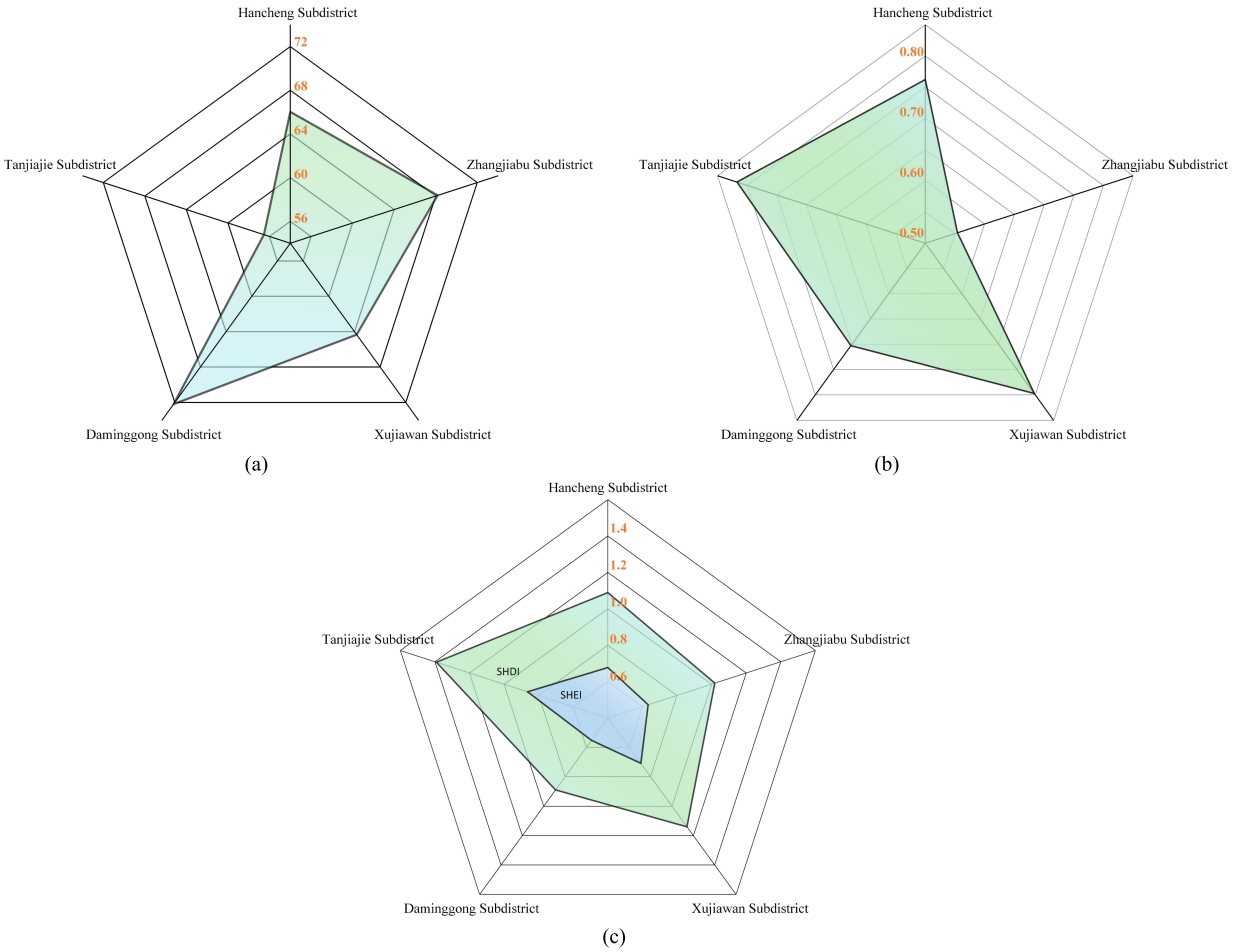

Fig. 19. Urban function pattern index results. (a) CONTAG. (b) DIVISION. (c) SHDI and SHEI.

All five subdistricts contain the corresponding fundamental infrastructures, such as campus; however, when analyzing the spatial rationality for such, PLAND is considered less helpful. Therefore, as shown in Fig. 21, the functional service index is involved in order to solve this problem.

The spatial pattern of functional zones can be manifested by the MNN index. As shown in Fig. 18(b), the commercial district shows higher MNN, indicating that there might be a distance between each district, resulting in dissociation in spatial patterns. The commercial district shows higher MNN, suggesting that there might be a distance between each zone, which results in dissociation in spatial patterns. The residential districts in Xujiawan show the lowest MNN, which indicates that the residential districts are reunited. The residential district that obtains the highest MNN is located in the Hancheng subdistrict, with two zones separately locate. In this case, proposing real estate development in a nearby region is recommended.

AWMPFD is a measurement of spatial shape complexity. Generally, the more human-activity is involved in a certain functional zone, the higher the patch fractal dimension is, that is to say, the AWMPFD value would be higher. The AWNPFDs of commercial districts and residential districts are both relatively high in Xujiawan and Zhangjiabu, suggesting that the urbanization progress of the study area is developed from the center to the two-way direction.

2) Allocation Analysis of Different Functional Zones: To evaluate whether an urban is well-developed or not depends on multiple aspects, such as whether it is equipped with complete fundamental facilities, and whether the functional zones have played their respective role. For example, the service scope of the campus is considered to cover each residential district, while the functional zone spatial pattern of each subdistrict should be reasonable. In order to better analyze the rationality of urban functions, taking the purpose of building the spatial connection of different functional zones, the indicators for analyzing the connection between functional zones were proposed (see Fig. 19).

Tanjiajie subdistrict achieves low CONTAG index and high DIVISION index, indicating that connectivity was poor and fragmentation was high among functional zones, which further suggests that Tanjiajie maintains a relatively high urbanization level in the study area. SHDI represents the diversity of functional zone categories, while SHEI illustrates the uniformity of number and area for urban functional zones. Here, the SHDI and SHEI are high, indicating that the subdistrict might be more balanced in functional zones distributions, for example, in Fig. 19, the Tanjiajie and Xujiawan subdistricts. The urban functional zone could play a better part in human society

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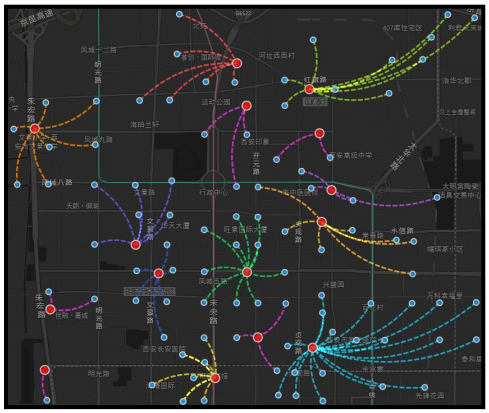

Fig. 20. Search the nearest campus in each residential district (the red points represent campuses, while the blue points represent residential districts).

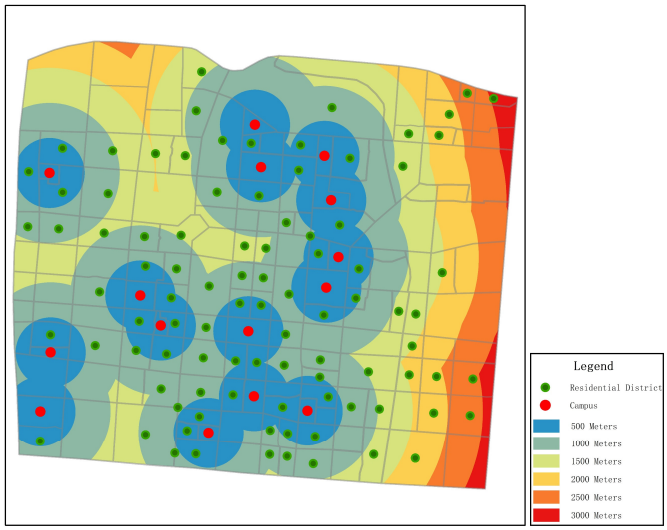

Fig. 21. Service scope of the campus.

by interrelating and interacting with each other. To evaluate the connection between each zone, the service scope evaluation index is proposed and tested by analyzing campus and residential districts as an example. Fig. 20 shows the result of the nearest campus location searching within each settlement based on the Euclidean distance. As can be seen from the diagram, the school facilities in the southeast study area appeared to be overloaded, yet, in the southwest, there is much looser; therefore, the campus allocation in the direction southeast and southwest of the study area is considered less reasonable.

Fig. 21 shows the functional service scope of primary and secondary schools. In total, there are 92 residential districts and 15 campuses in the study area. The service scope for the campus was set, respectively, at the scale of 500, 1000, 1500, and 2000 m. Within the 500 m, service scope locates 32 residential districts, accounting for 35% in total; when the parameter goes up to 1000, the number of residential districts rises as well, with a total of 34 and accounts for 37%. However, as the parameter further goes up to 1500, the number of residential districts decreases, with a total of 16, and accounts for 17%. Except for these, there are ten residential districts accounting for 10%, located beyond

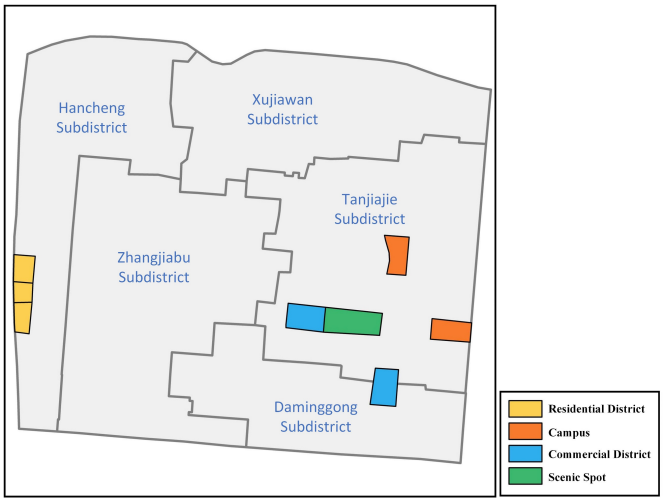

Fig. 22. Regional planning for unused land.

the 1500-m service scope, where the service radius is too large against the national standards, and the locations of such are all to east of the study area. This suggests that the service radius of primary and secondary schools in the eastern study area is relatively large, which surpasses 1500 m. Thus, the accessibility of schools is not so well as it is supposed to be.

3) Suggestions for Urban Function Planning: The ultimate goal of functional allocation analysis is to propose suggestions for urban functional planning. According to the results of the allocation analysis, advices for unused land-use planning are given in Fig. 22. In the south of the Hancheng subdistrict, residential districts were discretely distributed, and the school infrastructure is complete. Therefore, it is necessary to build up more residential districts on the unutilized land, gradually. The results demonstrated that the Daminggong subdistrict has mainly the high level of dominance by residential districts. It would be beneficial to build up a series of commercial districts, as it contains a great potential for commercial consumptions. In the east of the study area, especially in the Tanjiajie subdistrict, the spatial accessibility of schools is low; for this reason, public infrastructure, such as schools and scenic spots, is recommended to be built in the future.

V. Discussion

A. Effectiveness of Equal-Area Division Algorithm in OCNN Classification

To testify the validity of the equal-area division algorithm, the commonly used random point generation algorithm is used as the comparison method. Combining the two algorithms with OCNN, the confusion matrix calculated is shown in Fig. 23. Fig. 24 shows the local classification results of the remaining six major land cover types except for water bodies (because of the classification accuracies of both two-convolution kernel position generation methods). It can be seen that the former has relatively better recognition of the three types of land covers: buildings with blue roofs, artificial turf fields, and bare land. Fig. 23 also shows that the proposed equal-area division algorithm as well performs better for the other three

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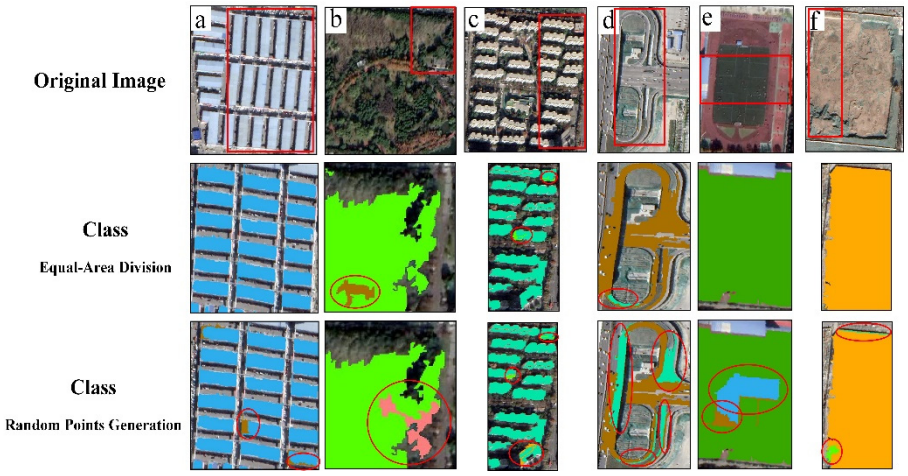

Fig. 23. Local classification results, respectively, by using equal-area division algorithm and random points' generation algorithm to generate convolution kernel position. (a) Building with the blue roof. (b) Grass. (c) Other building. (d) Road. (e) Artificial turf field. (f) Bare soil.

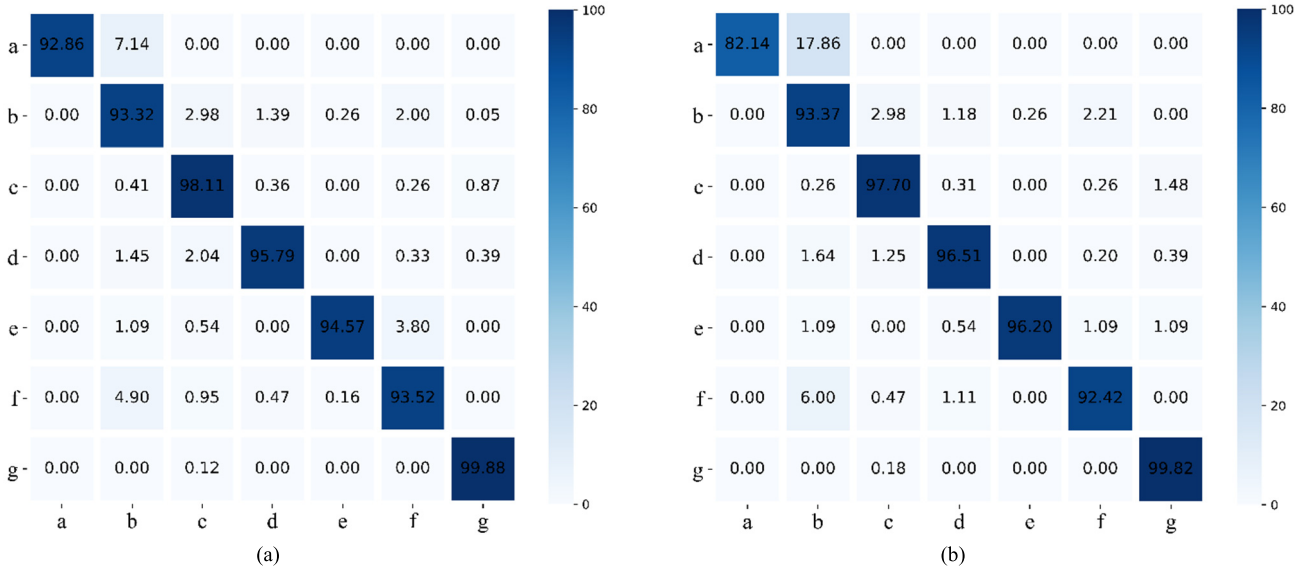

Fig. 24. Confusion matrix. (a: building with the blue roof; b: grass; c: other building; d: road; e: artificial turf field; f: bare soil; and g: water). (a) Equal-area division algorithm. (b) Random point generation algorithm.

types although wrong classifications still happen. Since the convolution positions of buildings are almost generated at the boundary, CNN will irrationally extract the object features of the surrounding land covers, which causes misclassification of buildings.

B. Performance of DW-GAM for Functional Zone Attribute Identification

The spatial distribution of urban function forms the spatial structure of the city and the spatial arrangement of various urban land use types. For example, most residential districts are formed by neatly arranged buildings and greenbelts; industrial areas consist of some office buildings, factory buildings, and vegetation. Thus, this article proposes to identify urban functional zones by exploring different arrangements of various land covers. The DW-GAM provides the possibility to achieve this goal. DW-GAM can deeply mine the spatial relationships of different types of nodes being expressed by graphs and then judge the functional zone attributes of each graph through the similarity of graph features.

Traditional methods for identifying functional zone attributes mostly use social perception data, such as POI to give the functional semantic information, or use CNN to explore the influential features in different functional zones. Therefore, to verify the performance of DW-GAM on exploring the spatial distribution characteristics of land covers to identify urban functional zones, the urban functional zone identification results, respectively, by DW-GAM, traditional POI voting method, and CNN classification method are compared. For the sake of keeping the variables the same, we use the block units of each level obtained by the method proposed in Fig. 16 as the functional zone boundary constraints. The overall accuracies of the POI voting method and CNN method are 81.25% and 79.50%, respectively. The comparison result is illustrated in Figs. 16 and 25.

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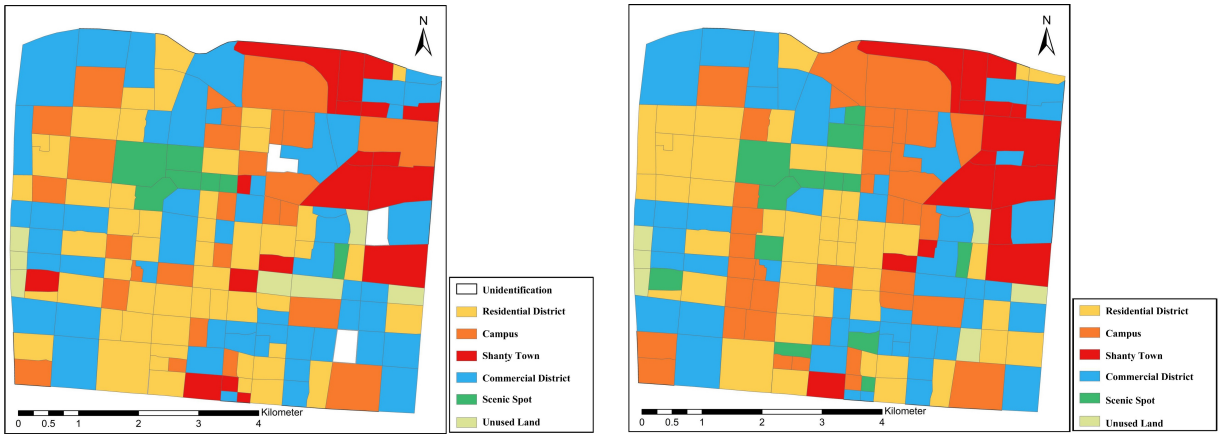

Fig. 25. Urban functional zone identification results, respectively, by (a) POI voting method and (b) CNN classification method.

From Fig. 25, it can be seen that the main problems of the POI voting-based method are given as follows: first, there is no POI in some blocks, which makes the blocks unclassified; second, although Shantytown and Scenic Spot can be recognized with high accuracy, the misrecognitions for other functional zones are a bit serious. In addition, the CNN classification method for identifying functional zones is neither ideal, especially the campuses and residential districts are severely misclassified, because CNN only focuses on deep features in fixed-size windows rather than the features of topological relations existing in the window scenes. By contrast, the proposed DW-GAM method performs best since it comprehensively considers both land cover types and their spatial topological relations.

C. Effectiveness of Multiscale Division

In fact, the boundary of each urban functional zone is not uniform. Some functional zones with large functional radiation areas may be extracted by large-scale block constraints. However, some large-scale blocks containing complex spatial distribution of diverse land covers should be divided on a fine scale by small-scale block constraints to further determine the attributes of urban functional zones on a fine scale.

Because the types of urban functional zones are complex and the size of block units are various, we cannot determine which scale of blocks should be optimally used for dividing urban functional zone. If the scale is too large, it may lead to underdivision, and the functional zones in the block cannot be fully extracted; if the scale is too small, overdivision will happen, which destroys the spatial structure of land covers in the functional zone and causes the redundancy of the sample size and sick graph feature expressing within the block.

Fig. 26 demonstrates the functional zone division results, respectively, by single-scale and multiscale methods. The proposed multiscale method divides a total number of 170 urban functional zone units. If this method is not employed, 299 urban functional zone units will be divided by a single scale method (using a small-scale road network to constrain block division). From the perspective of data amount to be identified, the division unit is reduced by 40%, and the time consuming for functional zone division and attribute identification is saved. Therefore, for the analysis of urban functional structure, the proposed method is more advantageous. Not only can the fine-scale division unit be derived from large scale but also the functional structure of the city can be reflected from a multiscale perspective.

Fig. 26. Functional zone division results, respectively, by (a) single-scale and (b) multiscale methods.

VI. Conclusion

Urban functional zone identification and functional pattern analysis are of great significance for urban planning and urban sustainable development. By combining using VHSR image, POI data, and multiscale road network data, this article proposes a novel multiscale recursive model that incorporates POI FD analysis and DW-GAM graph classification to classify urban functional zone. Especially, this article presents a case study of urban functional zone identification and allocation analysis using a self-established urban functional pattern evaluation index. The experiments and comparisons show the feasibility and superior of the proposed method. The contributions of this work can be summarized as follows.

1) An equal-area division algorithm is proposed to generate more reasonable convolution kernel locations for OCNN, which guarantees the accuracy of land cover classification to a certain extent and further ensures the accuracy of functional zone identification.

2) Considering that multiple land covers with specific spatial relationships compose different functional zones, DW-GAM is originally proposed for urban functional zone classification.

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Rather than the CNN model, the DW-GAM is used to analyze spatial relationship features between-land cover variation in urban functional zone identification, which enhances the stringency of functional zone identification.

3) Using a multilevel road network to decompose block unit step by step, a recursive model for multiscale division of urban functional zones is proposed by combining POI and DW-GAM to realize the fine extraction of functional zones from top to bottom, which avoids the imperfection of using single road network (such as overdivision or underdivision of some blocks). In addition, the identification results by jointly using DW-GAM and CR can relieve the negative influence when there is no POI in some blocks or the POI data are inaccurate.

4) As an applicable goal of this work, from the perspective of functional pattern analysis, an evaluation index system of urban functional reasonable allocation is put forward, by which suggestions for urban functional planning can be given, which is practically meaningful for enriching the research related to urban functional zone issues.

In addition, there are still some concerns as follows for future research.

1) The method proposed in this article has some shortcomings in the functional zone attribute identification. Since the spatial structure difference between the land covers in industrial districts and commercial districts is not conspicuous enough, the model cannot divide the two functional zones accurately. In the future, more social perception data, such as GPS trajectory data, can be introduced to explore human mobility among different functional zones to further distinguish different types of urban functional zones. Besides, aiming at better distinguishing the two types of functional zone, other network models based on graph classification can also be introduced to more accurately express the spatial relation features within the commercial districts and industrial districts.

2) The allocation of the urban functional pattern may affect the utilization of resources and the social cohesion of people and society, which further determines whether the urban development is sustainable. This article presents a primary framework for urban functional allocation analysis from the perspective of functional zone spatial patterns. However, since urban function planning involves many factors, such as human, resources, and environment, it is necessary to introduce more factors to analyze the urban functional allocation. For example, the changing pattern of urban population can be extracted from nighttime light data, and urban housing price data can be used to reflect the urban economy. In addition, the multiscale functional pattern analysis theory and indicators system should be established in the future to more effectively express the structural characteristics of functional zones so that the relevant research results can better serve the geographical pattern monitoring and urban intelligent management.

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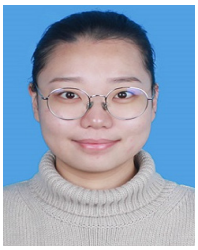

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